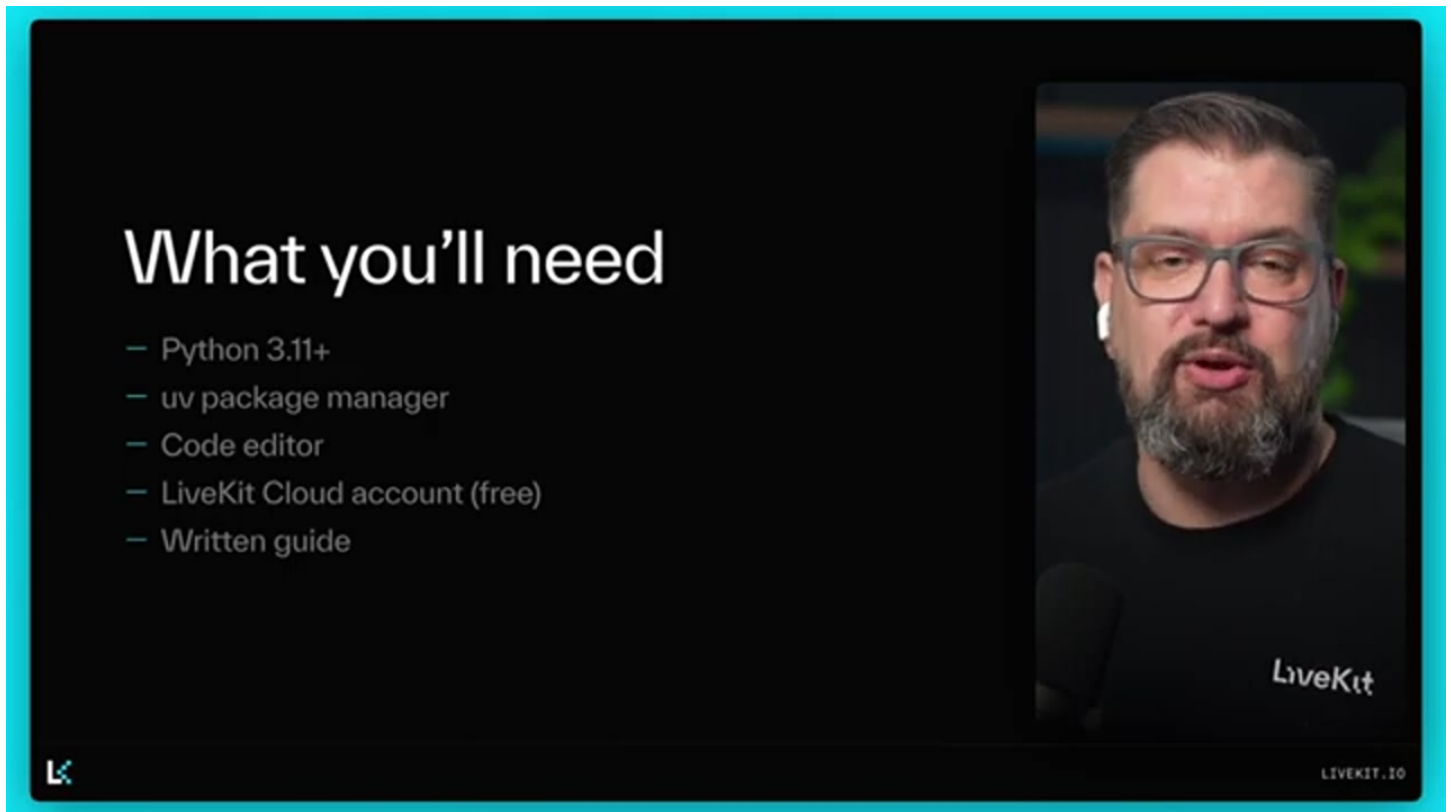


Lesson 1: Voice Agent Foundations

Section: LiveKit Voice AI Course

Summary: Coverage of Lesson 1: Voice Agent Foundations (03:03 - 13:24)

■ [03:03 - 03:23]



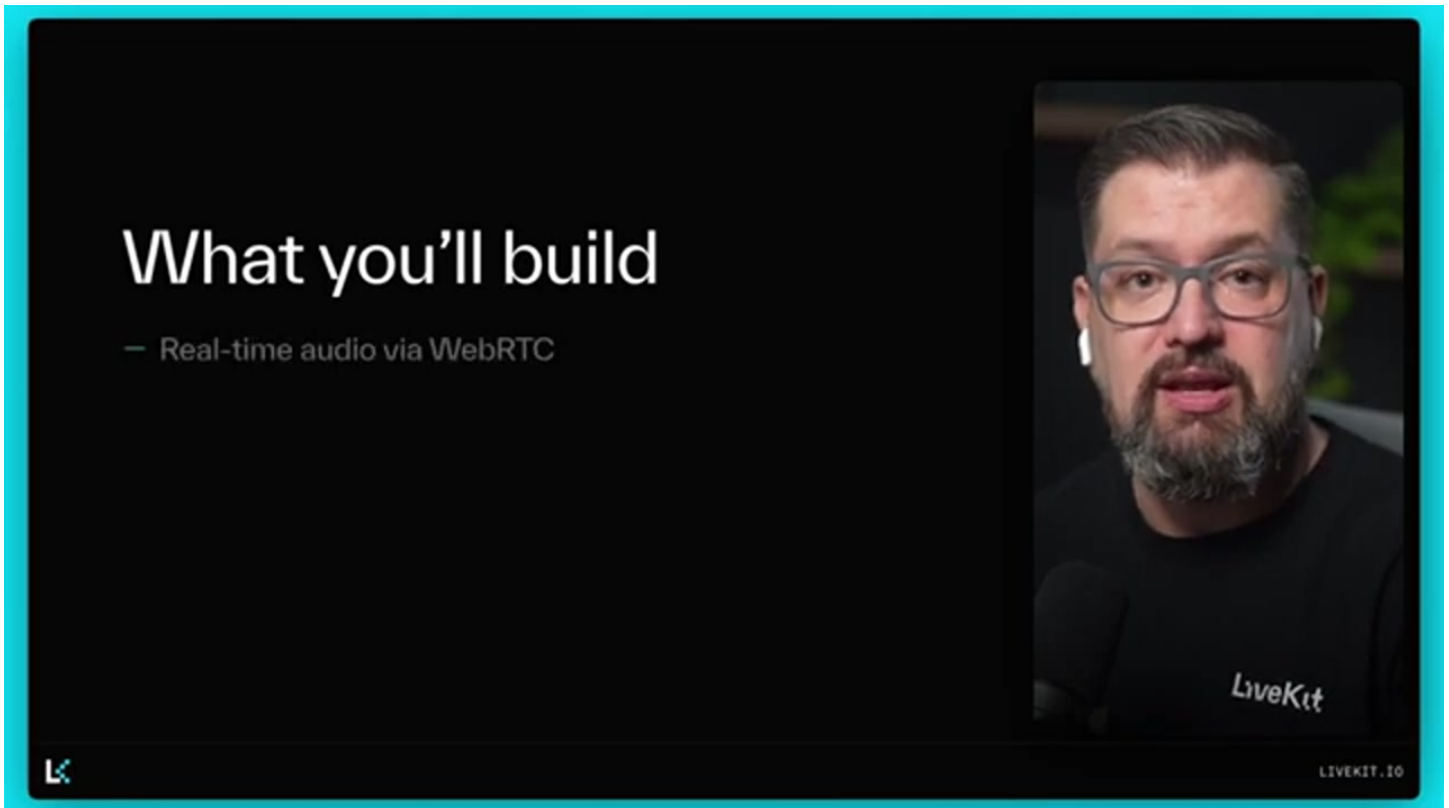
The video frame shows a presentation slide with a black background and white text. The title "What you'll need" is in a large, bold font. Below it is a bulleted list of requirements. In the top right corner, there is a small video inset of a man with a beard and glasses, wearing a black t-shirt with the "LiveKit" logo. The bottom left corner of the slide features a small "LK" logo, and the bottom right corner shows the URL "LIVEKIT.IO".

What you'll need

- Python 3.11+
- uv package manager
- Code editor
- LiveKit Cloud account (free)
- Written guide

LK LIVEKIT.IO

materials. So, got all that? Let's start building something. Before we write any code, let's build a mental model of how voice agents actually work. And by the end of this lesson, you'll understand which components make up a real-time voice agent, how they interact, and why latency is such a big deal. Throughout this workshop, you'll build a production quality voice assistant. It listens to



quality voice assistant. It listens to callers in real time through LifeKit's Web RTC infrastructure. It streams speech to text, runs responses through an LLM, and converts them back to speech with texttospeech. You'll add interruption handling using voice activity detection and semantic turn detection. And you'll wire up tool calls, conversation escalation, and

What you'll build

- Real-time audio via WebRTC
- Streaming STT, LLM, TTS
- Interruption handling
- Tool calling & metrics



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calls, conversation escalation, and metric collection. Every voice agent follows the same high-level pipeline. There are four core components. Voice activity detection determines when the human is speaking. Without it, your agent wastes money uh processing silence and might interrupt the user mid-sentence. Speech

The voice pipeline

- VAD: detects speech
- STT: transcribes audio



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interrupt the user mid-sentence. Speech to text transcribes audio into tokens the LLM can understand. You'll want to choose models that support your target languages and accents. The large language model generates the reply, prompts, planning, and tool integrations all live here. And text to speech converts the LLM response into audio that feels natural and on brand.

The voice pipeline

- VAD: detects speech
- STT: transcribes audio
- LLM: generates reply
- TTS: speaks response



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that feels natural and on brand. Supporting components like background noise cancellation, noise suppression, and end of turn detectors help the agent behave like a good conversational partner. And you'll add several of these throughout the workshop. Now, the cascaded pipeline from ST to LLM to TTS shown here is currently the most

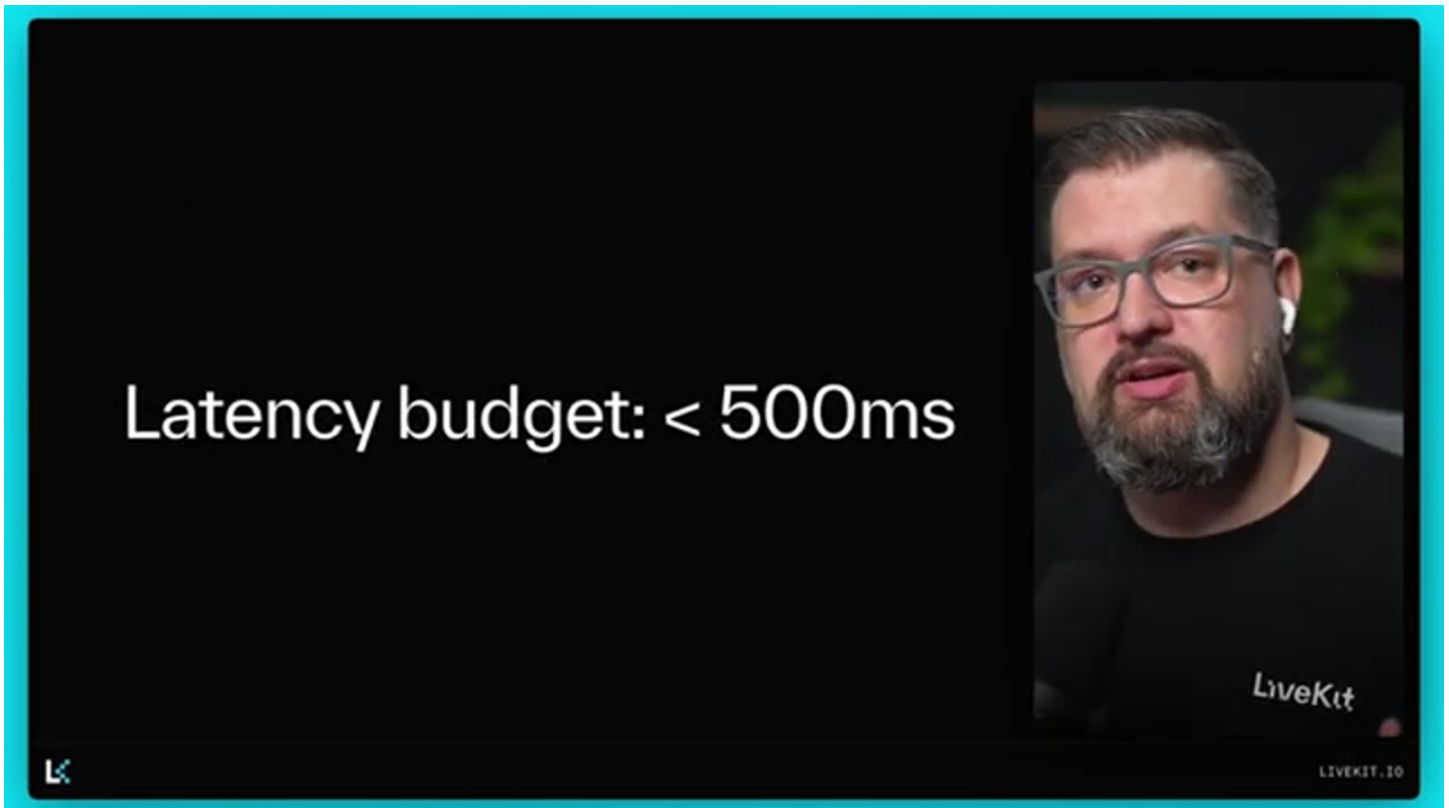
The voice pipeline

- VAD: detects speech
- STT: transcribes audio
- LLM: generates reply
- TTS: speaks response



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TTS shown here is currently the most popular approach. But there is an alternative and that is to use speech-to-text or real-time models that can process audio directly without explicit transcription. Now that's not going to be covered here in this workshop, but look out for more videos on that subject coming very soon. Now humans expect



coming very soon. Now humans expect conversational latency under 500 milliseconds. Every component in the pipeline adds delay. VAD is the fastest at 15 to 30 milliseconds. ST takes 200 to 600 milliseconds. Uh LLMs range from 100 milliseconds to a full second. TTS adds another 100 to 300 milliseconds.

Average latencies

Component	Best case	Typical
VAD	15-20ms	20-30ms
STT	200-300ms	400-600ms
LLM	100-200ms	500-1000ms
TTS	100-150ms	200-300ms
Total	~415ms	1.1s-2s



adds another 100 to 300 milliseconds. And so these all add up. And so you could be looking at anywhere from 400 milliseconds to 2 seconds total. Hey, how's it going? Good. How are you? I'm good. Just got back from lunch. What did you have? A burrito.

■ [05:43 - 06:03]



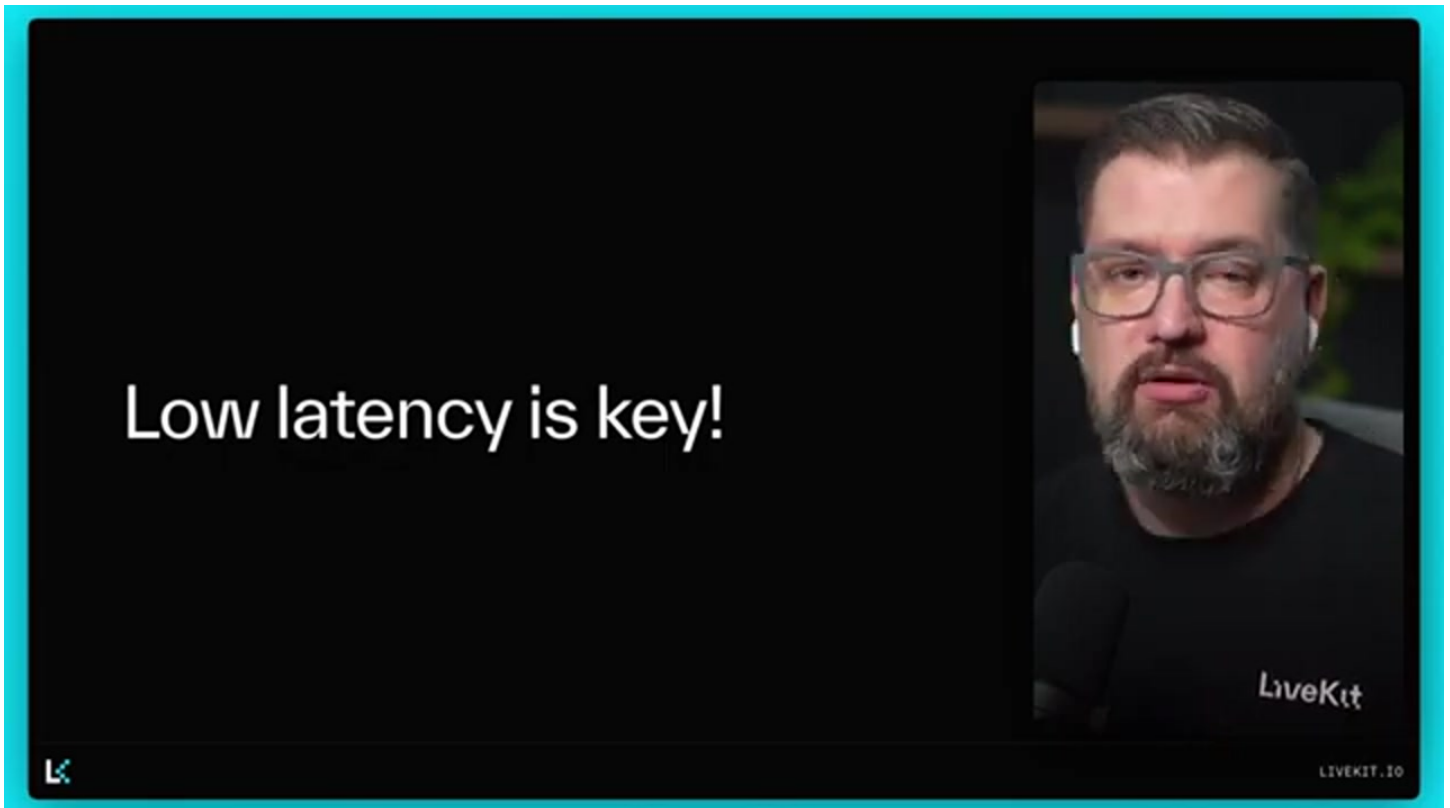
What did you have? A burrito. Anything else you'd like to talk about? You know what? I I got to go. That felt painful, right? Each 2C delay adds up

Low latency is key!



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painful, right? Each 2C delay adds up fast. In a real conversation, you'd probably give up after the second or third pause. Your users will do the same thing. Low latency agents keep pipelines streaming and parallel. They avoid blocking I/O and they carefully choose providers that balance quality with speed. And you'll learn techniques for



speed. And you'll learn techniques for shaving off those milliseconds in later lessons. A real-time voice has uh different networking needs than text. Let's compare the options. HTTP over TCP is great for text request responses, but it has a problem for live audio. You see, TCP guarantees that packets arrive

Why WebRTC?

— HTTP/TCP: head-of-line blocking



LIVEKIT.IO

see, TCP guarantees that packets arrive in order. If one packet gets lost, everything behind it has to wait until that packet is resent and received. That's called head of line blocking. For a web page, that's fine. For live speech, a 200 millisecond pause while waiting for a retransmit on a single packet ruins the entire experience. The

Why WebRTC?

— HTTP/TCP: head-of-line blocking



LIVEKIT.IO

packet ruins the entire experience. The other option is websockets. Web sockets are persistent and birectional, but they're still blocked by TCP retransmits under poor networking conditions. But then we have web RTC over UDP and which was designed for media. UDP doesn't guarantee packet order or delivery,

Why WebRTC?

- HTTP/TCP: head-of-line blocking
- WebSockets: still TCP retransmits
- WebRTC/UDP: skip lost packets



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guarantee packet order or delivery, which sounds bad until you realize that for live audio, skipping a lost packet is actually better than waiting for it. WebRTC builds smart handling on top of UDP. It uses Opus, a codec optimized for speech that compresses audio efficiently. Each packet carries a time stamp so the

WebRTC features

- Opus codec
- Packet timestamps



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Each packet carries a time stamp so the receiver knows exactly when to play it. A jitter buffer smooths out timing variations when packets arrive at uneven intervals. And an adaptive bit rate adjusts audio quality on the fly based on network conditions. All of this adds up to low latency even when the network isn't cooperating. So in practice, WebRTC lets your agent deliver first

WebRTC features

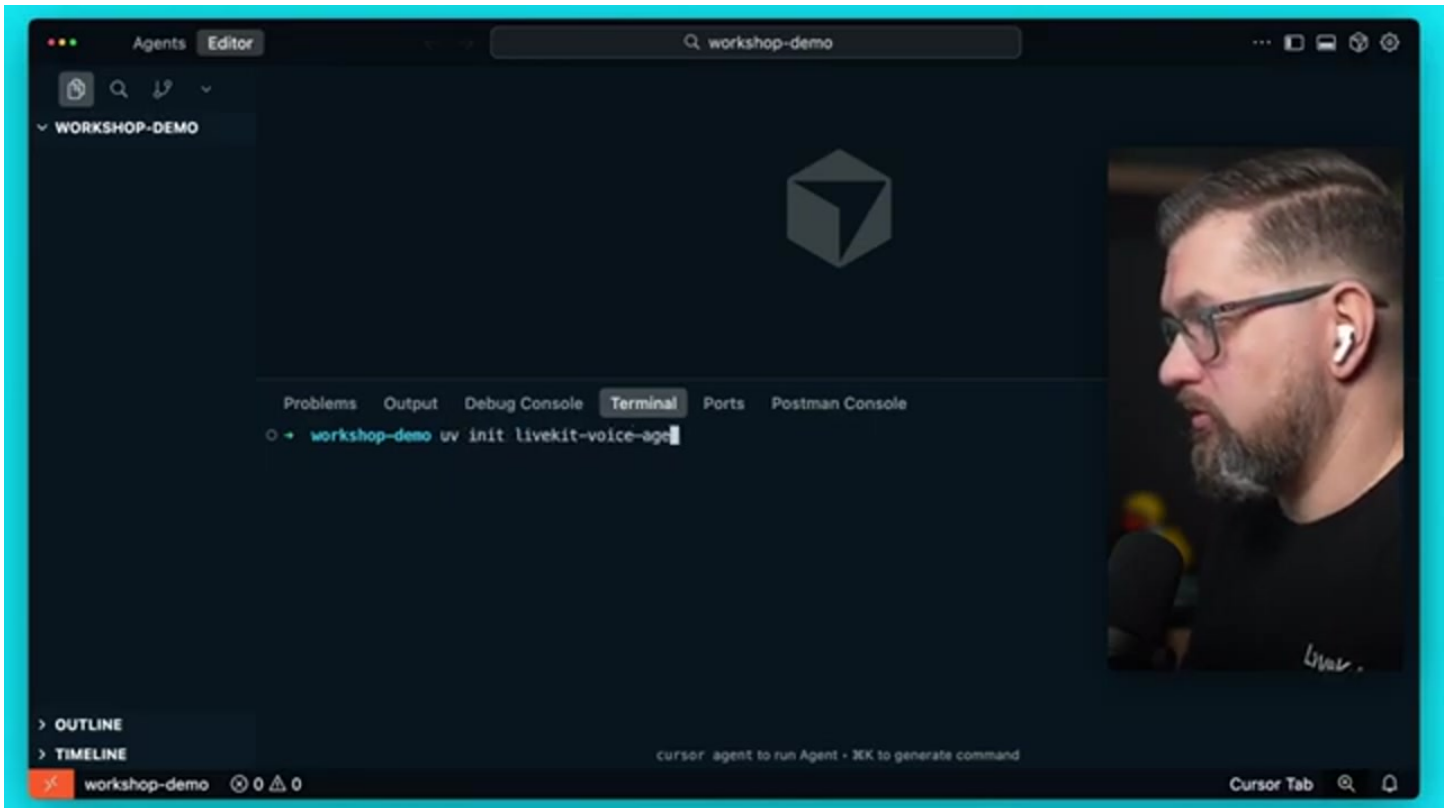
- Opus codec
- Packet timestamps
- Jitter buffer
- Adaptive bitrate



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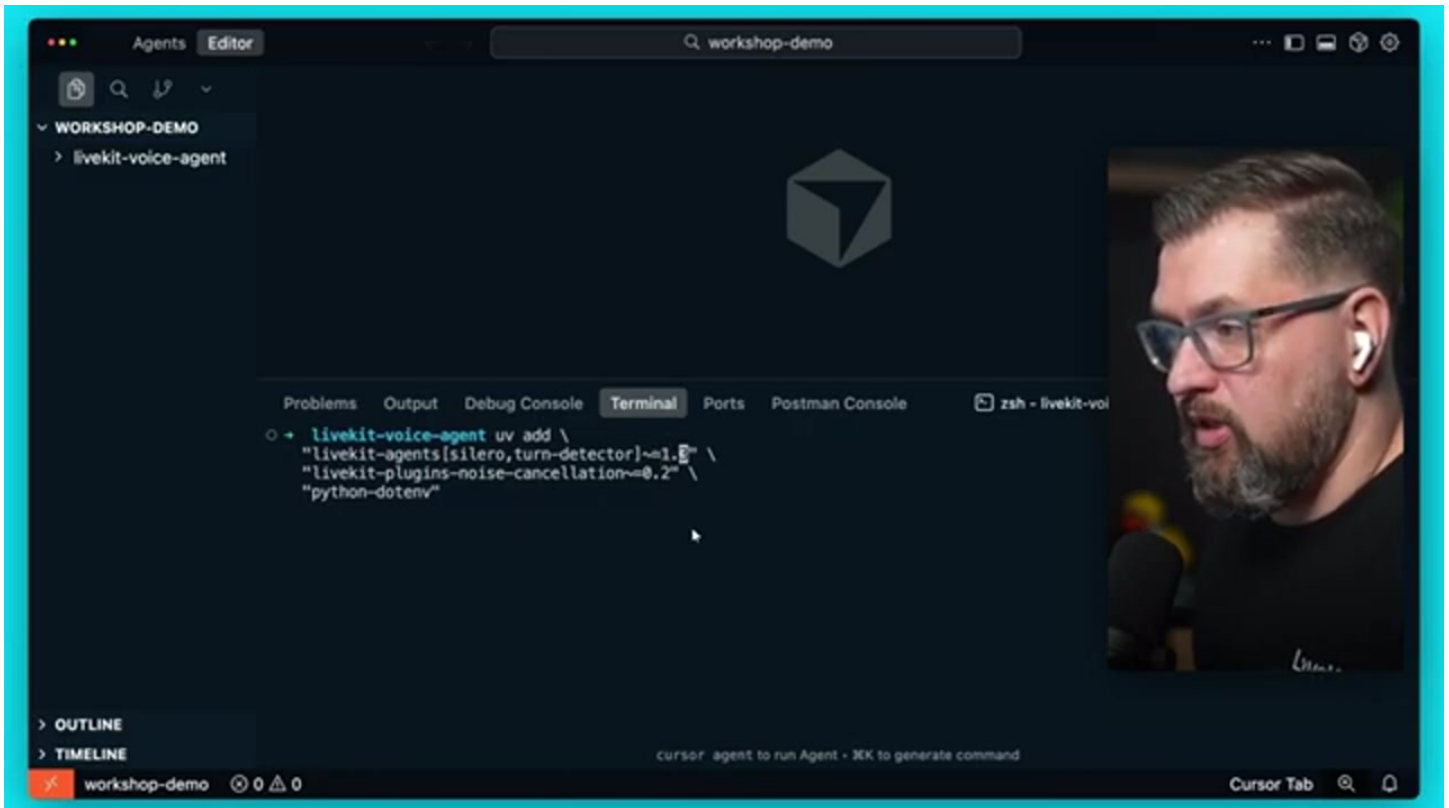
WebRTC lets your agent deliver first audio faster and stay responsive during packet loss and jitter. This workshop uses LiveKit's Web RTC infrastructure throughout. So let's set up a project. We're going to open up our terminal and initiate a new project using UV.

■ [08:23 - 08:43]



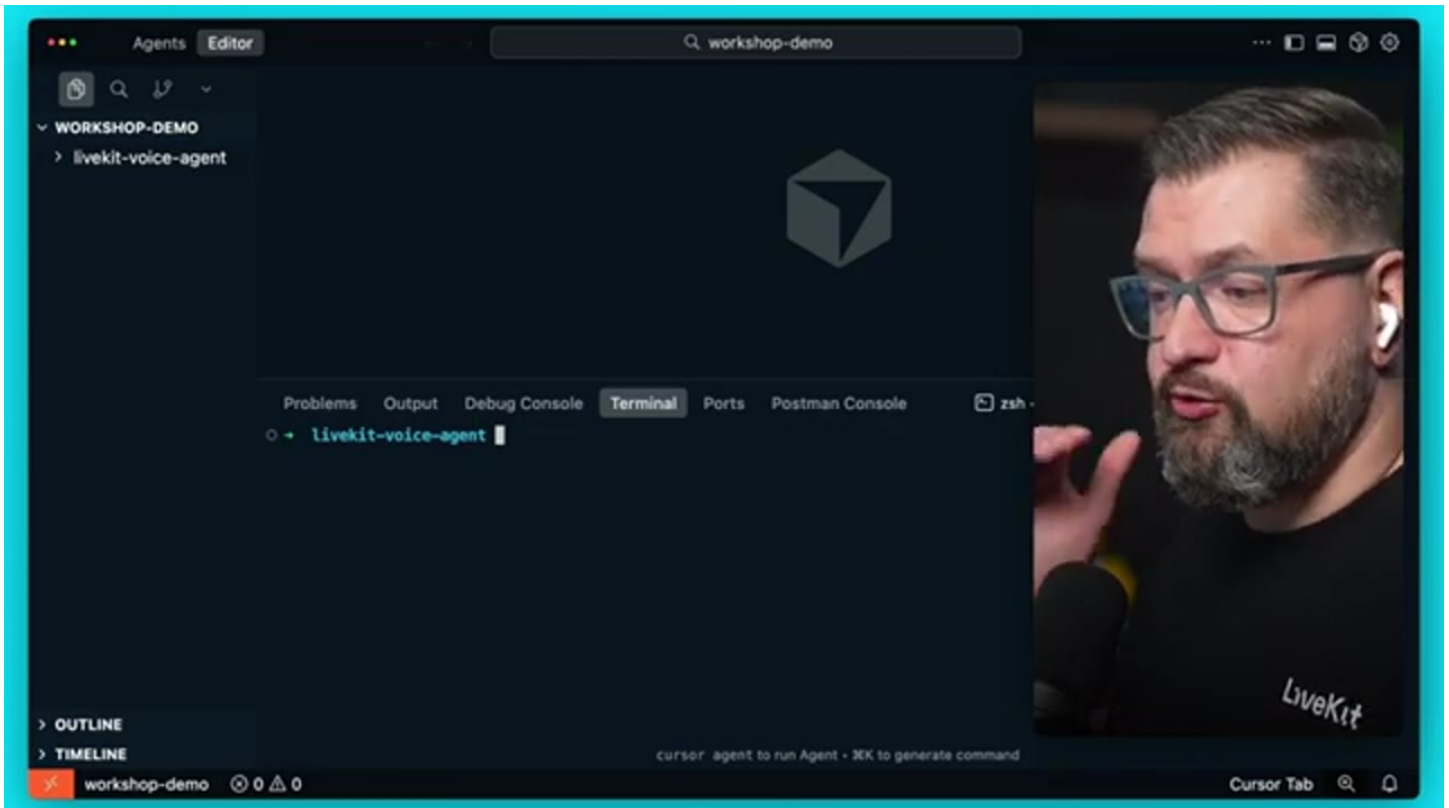
initiate a new project using UV. So the d-bear flag, it creates a minimal project without any boilerplate uh files. Next, we're going to install some dependencies. So UV add lifkit agents uh solo and turn detector. And it's good to note here we're using LifeKit agents uh version 1.3.

■ [08:43 - 09:03]



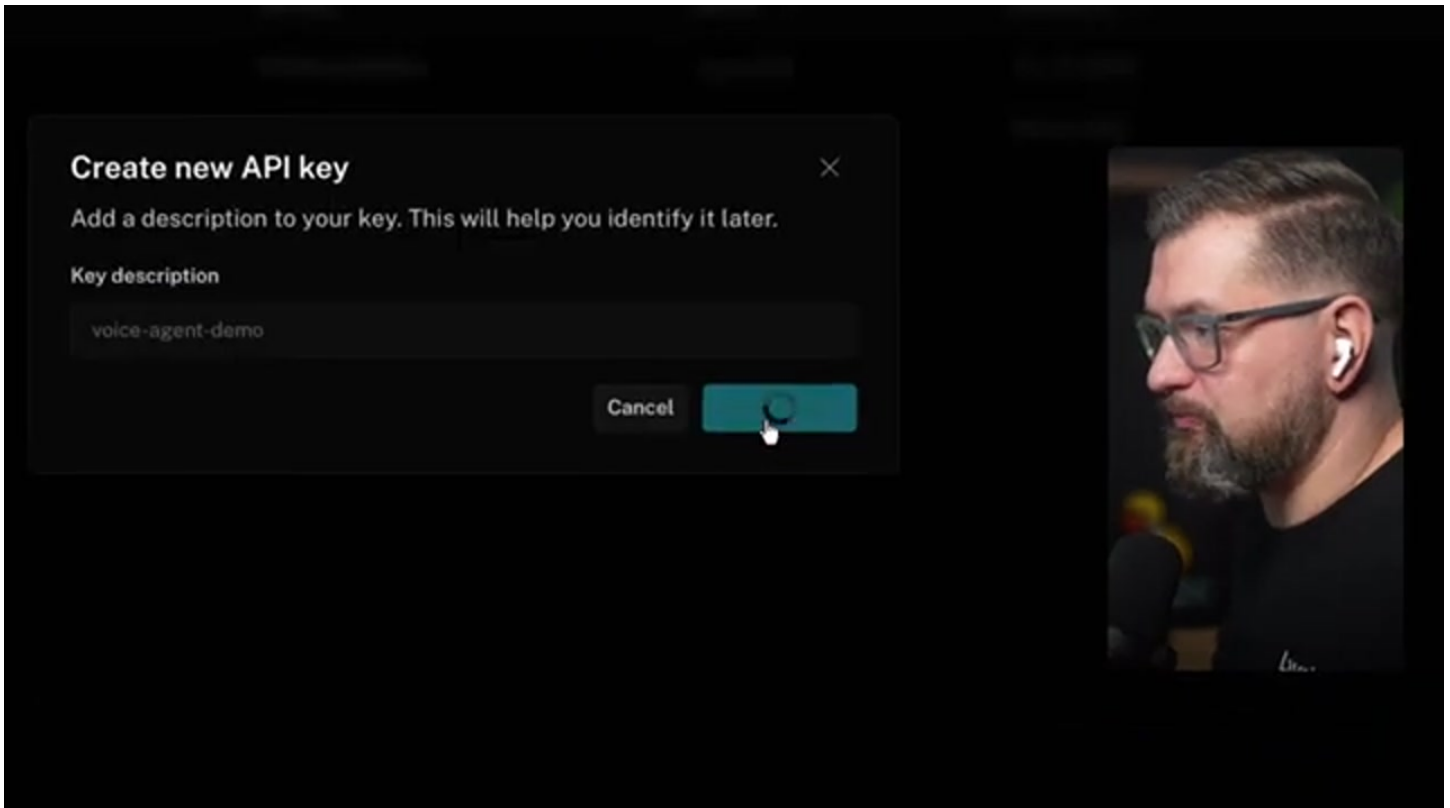
version 1.3. and then lifekit plugins noise cancellation and then python.vdv. So this installs the lifkit agents SDK with solar VAD and turn detection noise cancellation plugin and the python.v package for environment variables. Next, we need to go grab our uh API credentials from our LiveKit cloud

■ [09:03 - 09:23]



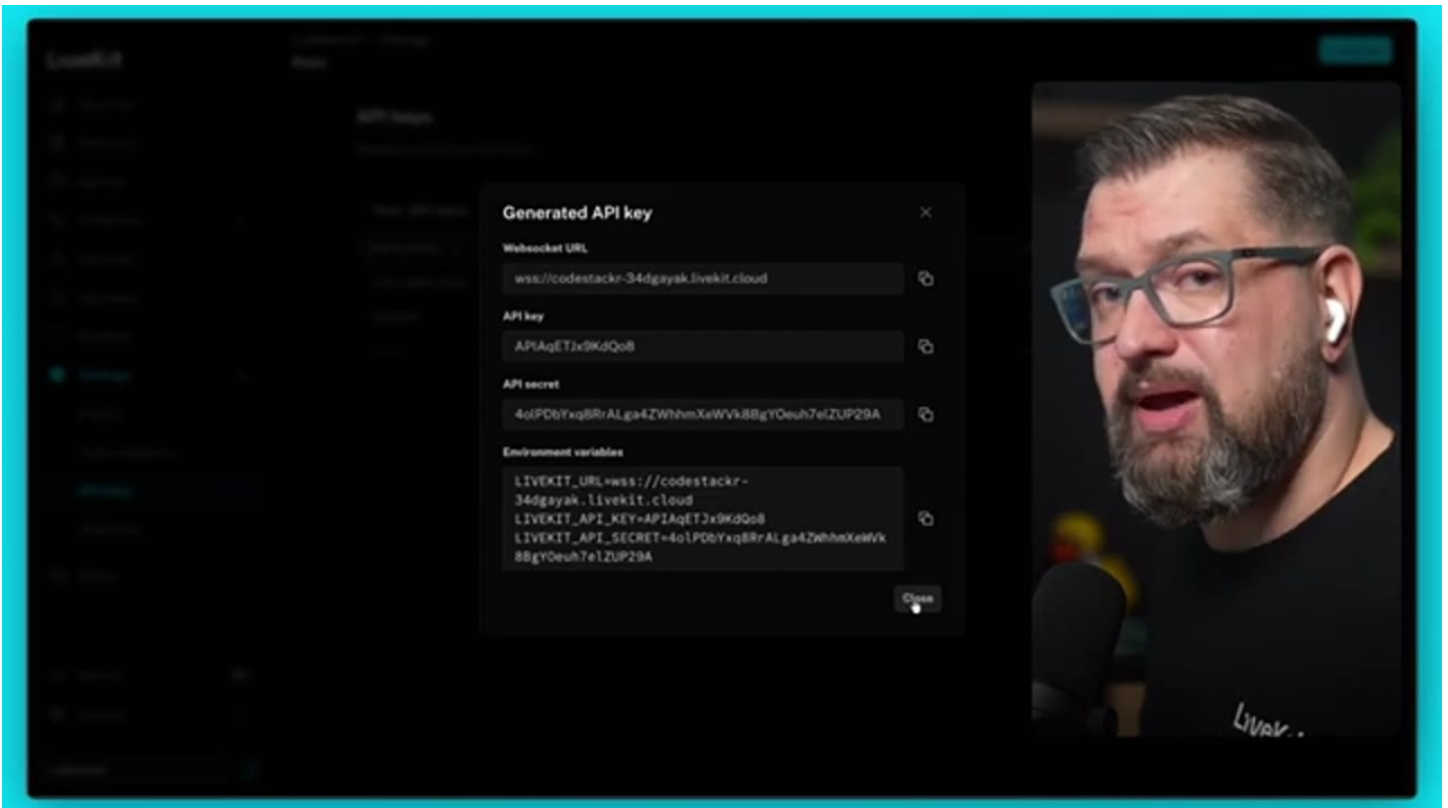
credentials from our LiveKit cloud dashboard. If you haven't created one yet, it's completely free. There's a link in the video description. So here on the dashboard, I'm going to go down to settings, API keys, and then I'm going to create a new key for this demo. And I'll just name this uh voice agent demo. And let's generate a new key.

■ [09:23 - 09:43]



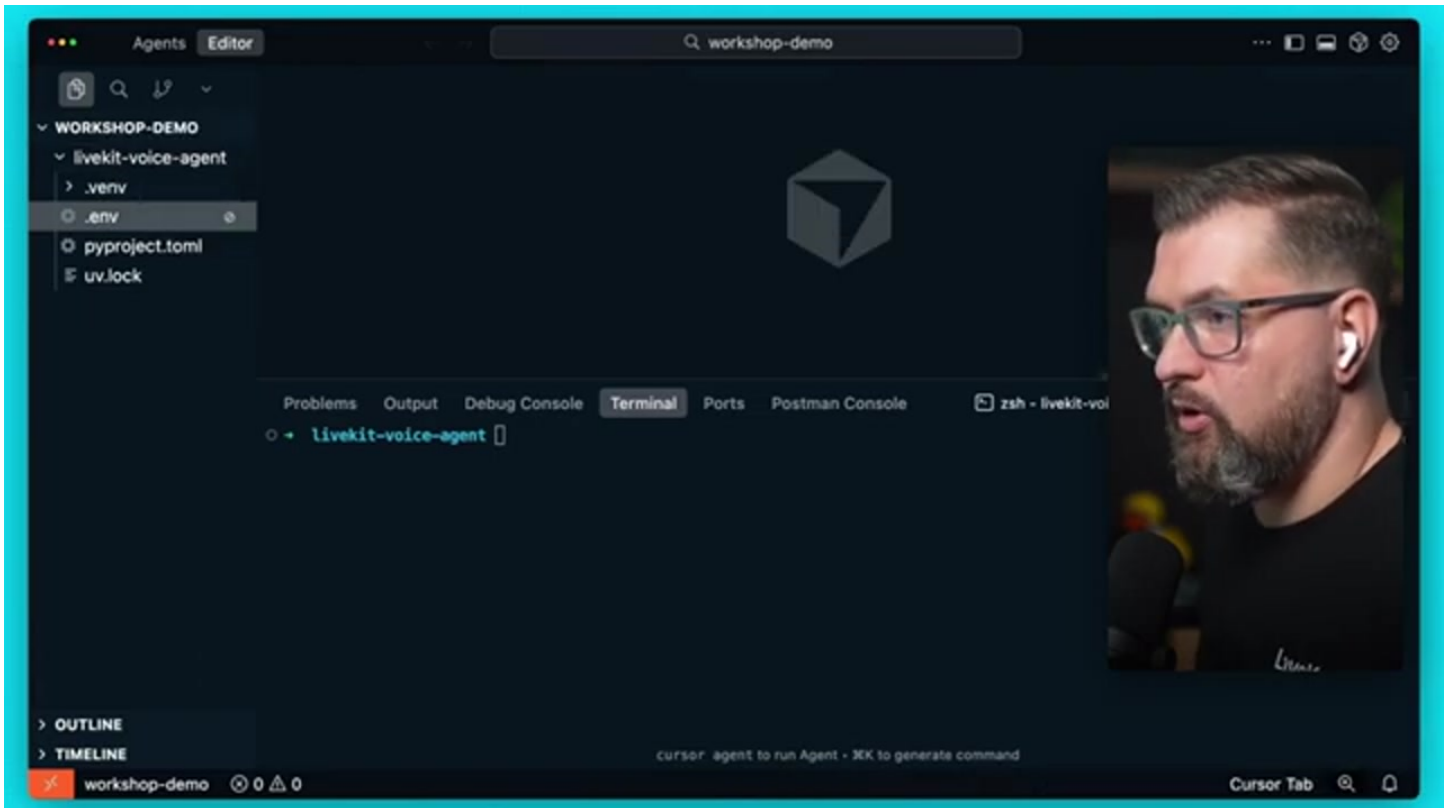
And let's generate a new key. So these are the things that you're going to need. The websocket URL, the API key, and the API secret. And to make things easier, you can just copy and paste uh these environment variables right here. And of course, I'm going to delete my API key. So, this one is not going to work for you. Uh, but again, it's completely free. Just go ahead and set one up for yourself. So, let's go

■ [09:43 - 10:03]



set one up for yourself. So, let's go back to the editor and we'll create a newv file. And we'll go ahead and paste in those credentials. Again, we have our LifeKit URL, our LifeKit API key, and our LifeKit API secret. So, make sure that you save that file, and then you can close it. And also make sure that the file is in your project directory. Okay, let's go ahead and create another

■ [10:03 - 10:23]



Okay, let's go ahead and create another file um again in our project directory here. We're going to call it agent. py. And so this is going to be our starter agent. And I'm going to paste in some code here and we'll walk through it. Now again, if you're following along in the written guide, you can copy and paste the code as well from there. So at the top here, we've got our imports from the

■ [10:23 - 10:43]

```
agent.py 4 X
livekit-voice-agent > agent.py > ...

1  import logging
2
3  from dotenv import load_dotenv
4  from livekit import agents
5  from livekit.agents import Agent, AgentServer, AgentSession, JobContext, room_io
6  from livekit.plugins import noise_cancellation, silero
7
8  load_dotenv()
9
10
11 class Assistant(Agent):
12     def __init__(self) -> None:
13         super().__init__(
14             instructions="You are a helpful voice AI assistant.",
15         )
16
17
18 server = AgentServer()
```



top here, we've got our imports from the LifeKit agents SDK. The agent class defines your agents behavior. The agent server handles dispatching sessions. And the agent session configures the voice pipeline. We have a job context here which provides information about the current job when a particular uh participant joins. And then in our session start we have uh the room IO

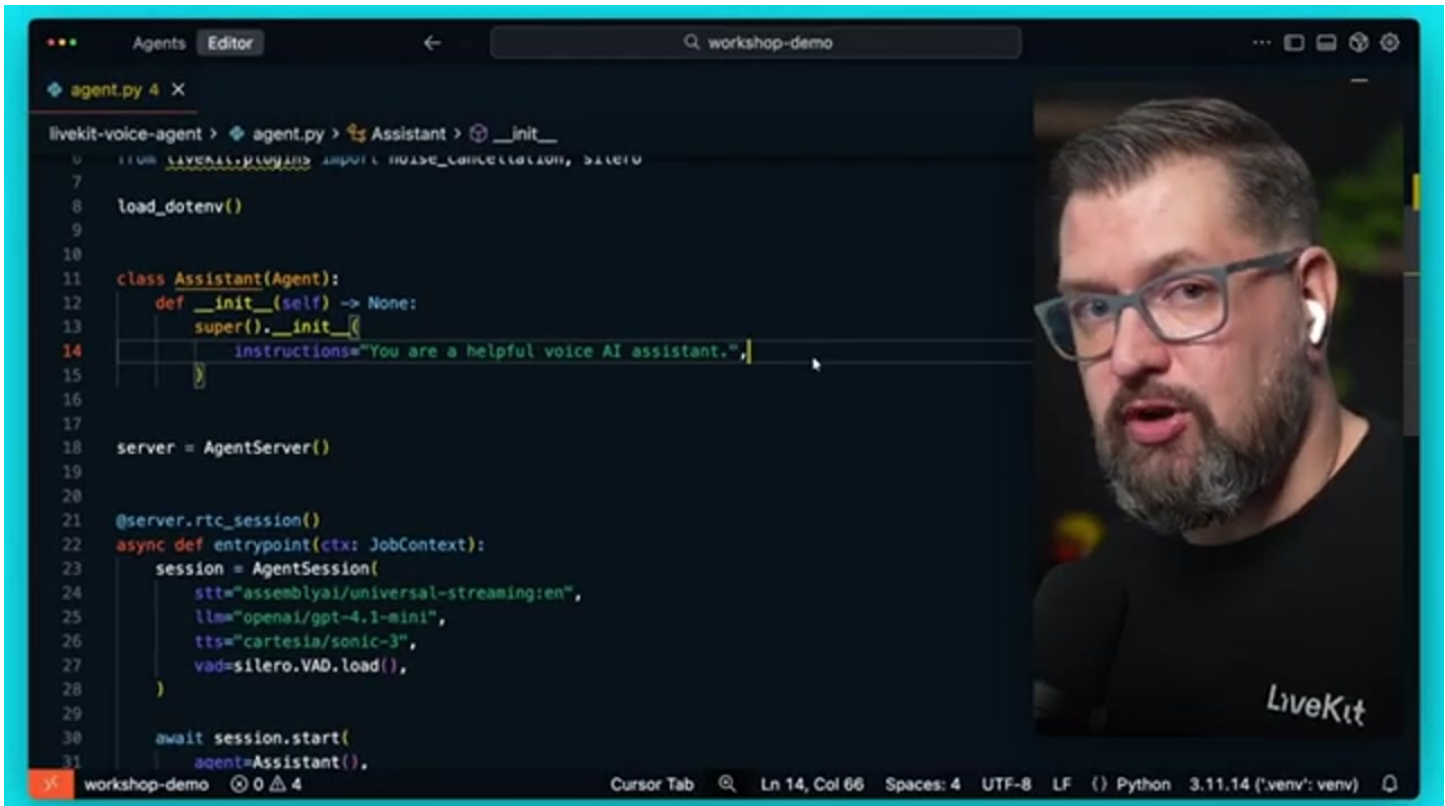
■ [10:43 - 11:03]

```
21 @server.rtc_session()
22 async def entrypoint(ctx: JobContext):
23     session = AgentSession(
24         stt="assemblyai/universal-streaming:en",
25         llm="openai/gpt-4.1-mini",
26         tts="cartesia/sonic-3",
27         vad=silero.VAD.load(),
28     )
29
30     await session.start(
31         agent=Assistant(),
32         room=ctx.room,
33         room_options=room_io.RoomOptions(
34             audio_input=room_io.AudioInputOptions(
35                 noise_cancellation=noise_cancellation.BVC(),
36             ),
37         ),
38     )
39
```



session start we have uh the room IO which provides options for configuring audio input and output. So in the agent session here you can see we're configuring our ST LLM and TTS and also our VAD. And just up here we have our basic instructions for this agent. you are a helpful voice AI agent. So very simple starter agent here for you. So

■ [11:03 - 11:23]



```
livekit-voice-agent > agent.py > Assistant > __init__
0 from livekit.plugins import noise_cancellation, silero
7
8 load_dotenv()
9
10
11 class Assistant(Agent):
12     def __init__(self) -> None:
13         super().__init__(
14             instructions="You are a helpful voice AI assistant.",
15         )
16
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18 server = AgentServer()
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21 @server rtc_session()
22 async def entrypoint(ctx: JobContext):
23     session = AgentSession(
24         stt="assemblyai/universal-streaming:en",
25         llm="openai/gpt-4.1-mini",
26         tts="cartesia/sonic-3",
27         vad=silero.VAD.load(),
28     )
29
30     await session.start(
31         agent=Assistant(),
```

workshop-demo 0 4 Cursor Tab Ln 14, Col 66 Spaces: 4 UTF-8 LF () Python 3.11.14 ('.venv': venv)

simple starter agent here for you. So let's go ahead and get this agent running and see what it does. So again, be sure that you have saved this file. Let's open up our terminal. So first thing that we need to do is download the required model files like the Solero VAD model. Now we only have to do this the very first time. So uvun uh agent.py and then download files. Now let's run the

■ [11:23 - 11:43]

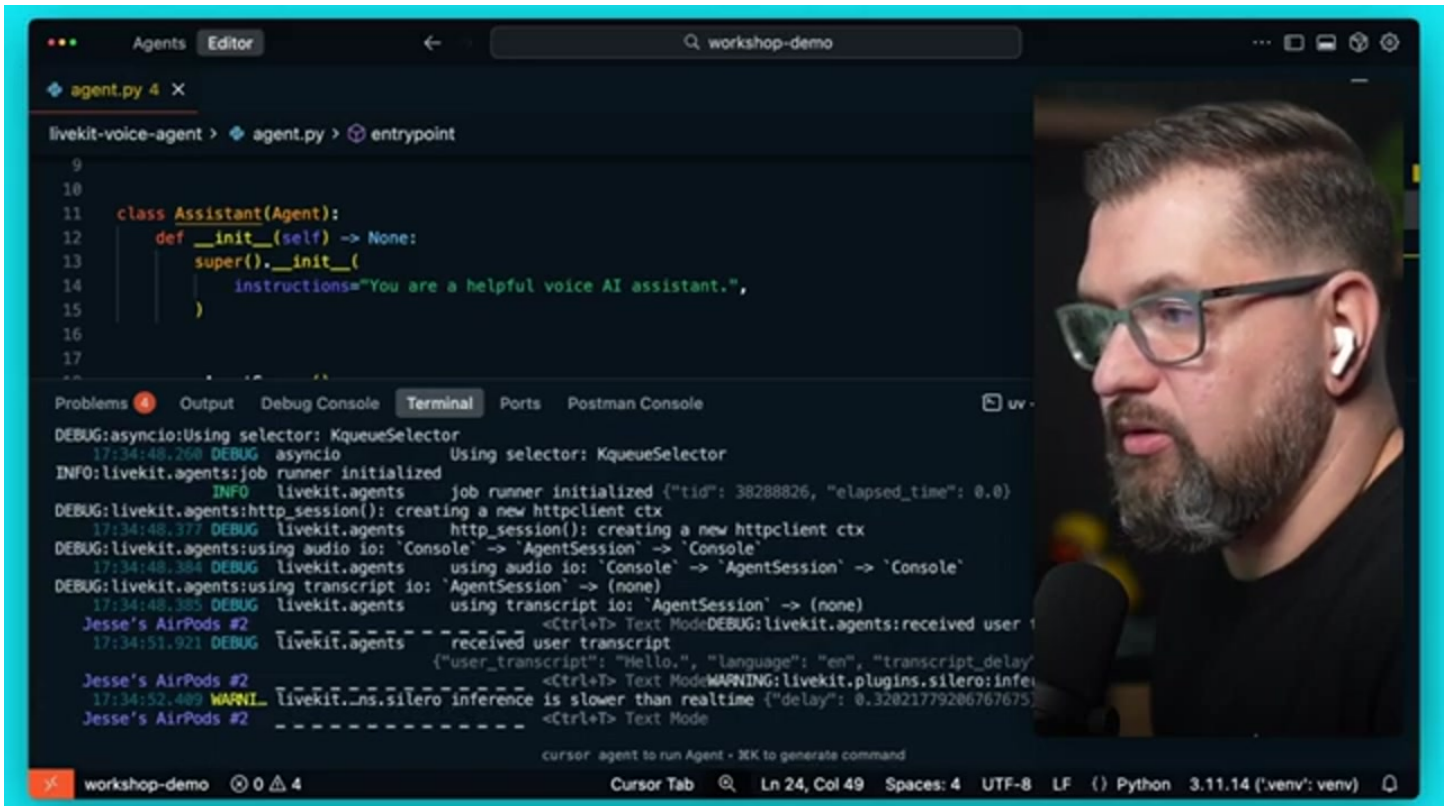
```
class Assistant(Agent):  
    def __init__(self) -> None:  
        super().__init__(  
            instructions="You are a helpful voice AI assistant.",  
        )
```

Output Debug Console Terminal Ports Postman Console zsh - livekit-vo
t-voice-agent uv run agent.py download



then download files. Now let's run the agent. So we can just do the same thing except remove the download files. So, u
agent and then console. So, we're going to actually run this right here in the console. Hello. >> Hello. How can I assist
you today? >> Oh, great. What What can you assist me with?

■ [11:43 - 12:03]



```
agent.py 4 X
livekit-voice-agent > agent.py > entrypoint

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11 class Assistant(Agent):
12     def __init__(self) -> None:
13         super().__init__(
14             instructions="You are a helpful voice AI assistant.",
15         )
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```
DEBUG:asyncio:Using selector: KqueueSelector
17:34:48.260 DEBUG asyncio Using selector: KqueueSelector
INFO:livekit.agents:job runner initialized
INFO livekit.agents job runner initialized {"tid": 38288826, "elapsed_time": 0.0}
DEBUG:livekit.agents:http_session(): creating a new httpclient ctx
17:34:48.377 DEBUG livekit.agents http_session(): creating a new httpclient ctx
DEBUG:livekit.agents:using audio io: 'Console' -> 'AgentSession' -> 'Console'
17:34:48.384 DEBUG livekit.agents using audio io: 'Console' -> 'AgentSession' -> 'Console'
DEBUG:livekit.agents:using transcript io: 'AgentSession' -> (none)
17:34:48.385 DEBUG livekit.agents using transcript io: 'AgentSession' -> (none)
17:34:48.385 DEBUG livekit.agents using transcript io: 'AgentSession' -> (none)
Jesse's AirPods #2 ----- <Ctrl+T> Text ModeDEBUG:livekit.agents:received user transcript
17:34:51.921 DEBUG livekit.agents received user transcript
{"user_transcript": "Hello.", "language": "en", "transcript_delay": 0.0}
Jesse's AirPods #2 ----- <Ctrl+T> Text ModeWARNING:livekit.plugins.silero:inference is slower than realtime {"delay": 0.328217792867675}
17:34:52.489 WARNING livekit.plugins.silero:inference is slower than realtime {"delay": 0.328217792867675}
Jesse's AirPods #2 ----- <Ctrl+T> Text Mode
```

cursor: agent to run Agent - KK to generate command

workshop-demo 0 4 Cursor Tab Ln 24, Col 49 Spaces: 4 UTF-8 LF () Python 3.11.14 ('.venv': venv)

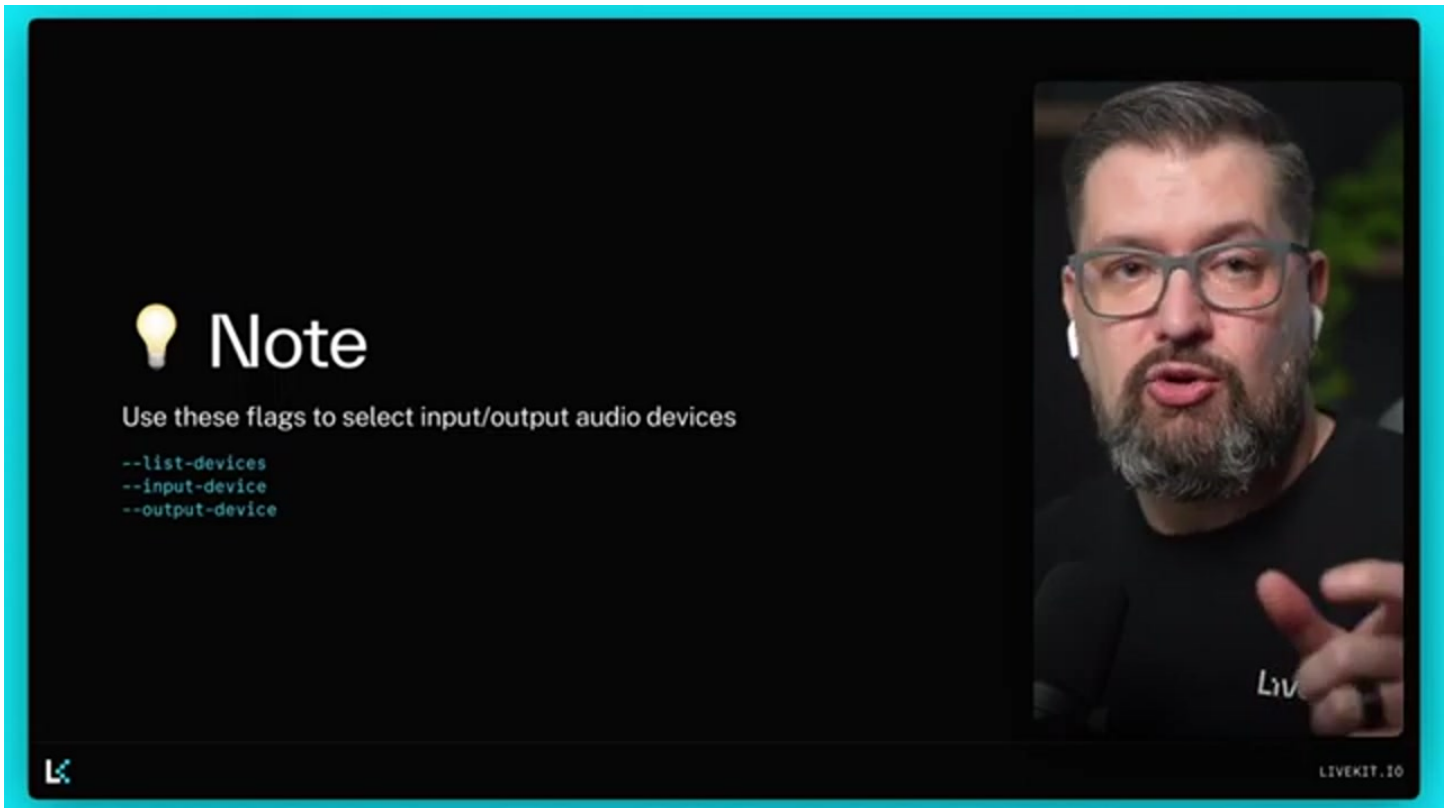
with? >> I can help with a variety of tasks such as answering questions, providing information, assisting with writing and editing, helping with learning new topics, generating ideas, offering recommendations, and much more. What do you need? >> Very helpful. Thank you. Thank you very much. We'll we'll talk to you again later. Bye-bye. Very helpful assistant. Very nice. Now, to stop your agent from running, just press control C. Now, if

■ [12:03 - 12:23]



```
Agents Editor workshop-demo
agent.py 4 X
livekit-voice-agent > agent.py > entrypoint
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11 class Assistant(Agent):
12     def __init__(self) -> None:
13         super().__init__(
14             instructions="You are a helpful voice AI assistant.",
15         )
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■ [12:23 - 12:43]



💡 **Note**

Use these flags to select input/output audio devices

```
--list-devices  
--input-device  
--output-device
```

LIVEKIT.IO

output devices using the input device and output device flag. Now, voice agents deal with messy real world inputs. People mumble. They change their minds midsentence, and they ask things in ways that you never anticipated. So, starting to think about how you'll test these scenarios now will save you headaches later. Now, testing and

■ [12:43 - 13:03]

Testing mindset



LiveKit

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headaches later. Now, testing and metrics, they come up later uh in a later lesson, but start keeping a backlog of scenarios that you want to test as features land. Uh so think about like edge cases, different accents, background noises and interruptions. So now you have a working voice agent

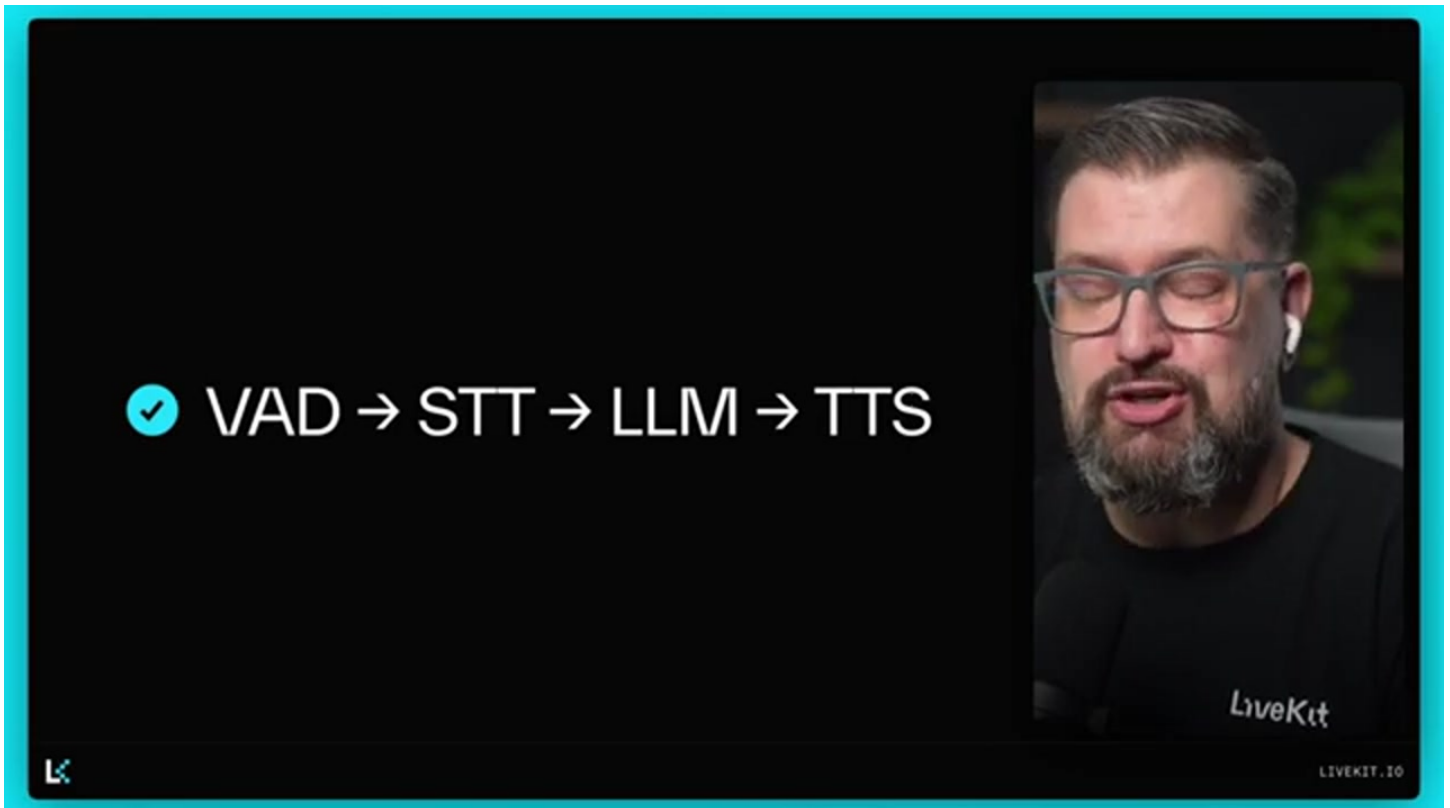
■ [13:03 - 13:23]

✓ VAD → STT → LLM → TTS



So now you have a working voice agent and you understand the core pipeline. The that VAD detects voice ST transcribes it. The LLM generates a response and TTS uh speaks it back. So, in the next lesson, you'll add semantic turn detection to prevent those awkward interruptions when an agent jumps in too

■ [13:23 - 13:24]



The image shows a video player interface with a black background. On the left, a presentation slide displays the text "✓ VAD → STT → LLM → TTS" in white. On the right, a video feed shows a man with a beard and glasses speaking into a microphone. The "LiveKit" logo is visible in the bottom right corner of the video feed. In the bottom left corner of the player, there is a small "K" logo. In the bottom right corner of the player, the text "LIVEKIT.IO" is displayed.

interruptions when an agent jumps in too early. Hey, I wanted to ask you um about